

Reproducing Transfer-Learning and Snapshot-Ensemble Diabetic Retinopathy Grading on APTOS 2019: Stage 2 (Base Paper & Reproducibility Check) and Stage 3 (Baseline Reproduction)

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CV Research-Reproduction Project — Diabetic Retinopathy Grading

Abstract—We select Chilukoti et al. (2024), which reports a quadratic weighted kappa (QWK) of 0.954 on APTOS 2019 with an ImageNet-initialised EfficientNet-B3 and 0.967 with EyePACS pre-training and snapshot ensembling, as our base paper, document its reproducibility, and re-implement it from the text because no code is public. We run the exact published protocol (300 px, 60 epochs, batch 16) on a Colab T4 GPU, after compute-limited pilot runs (224 px, 30 epochs) on an Apple M2. With the paper’s hyper-parameters the best-validation model reaches test QWK 0.858 and the 10-snapshot vote 0.846, about 0.10 below the reported 0.954. Training at $\text{lr } 10^{-3}$ is unstable at both resolutions, and Severe NPDR is almost never recognised (recall ≤ 0.10). Changing only the learning rate to 10^{-4} makes training stable and gives QWK 0.882 for the snapshot ensemble, with PDR recall rising to 0.57. The full resolution and schedule add about 0.03 QWK at the paper’s learning rate but nothing at 10^{-4} , so a gap of about 0.07 remains that none of the stated settings explain. We also find 123 groups of byte-identical images in APTOS (30 with conflicting grades), whose leakage into the test set has no measurable effect. No clinical claims are made.

I. Stage 2: Base Paper and Reproducibility Check

A. Selected base paper

We reproduce Chilukoti et al., “A reliable diabetic retinopathy grading via transfer learning and ensemble learning with quadratic weighted kappa metric,” BMC Medical Informatics and Decision Making, 24:37, 2024 [1]. The paper fine-tunes an ImageNet-pretrained EfficientNet-B3 with a three-layer classifier head, pre-processes fundus images with CLAHE followed by Gaussian blur, and ensembles predictions from weights saved at several epochs of a single training run (“snapshot” ensembling). It reports QWK as the primary metric: 0.954 on APTOS 2019 for a single model trained from ImageNet weights, and 0.967 when the model is first trained on EyePACS and the snapshot ensemble is used.

B. Why this paper

It matches the direction set by our literature review: it treats QWK (an ordinal, imbalance-aware agreement metric) as primary, uses a transfer-learning CNN that fits

free-tier hardware, evaluates on the public five-class APTOS 2019 dataset, and leaves the two gaps we identified untouched: no imbalance-aware loss and no ordinal-aware training objective. That makes it a clean base for a single-variable Stage 4 hypothesis.

C. Reproducibility checklist

TABLE I
Reproducibility checklist for the base paper.

Item	Our answer
Public code?	No. No repository is linked and none was found on GitHub; we re-implemented the method from the paper text (Sec. II).
Dataset public & size?	Yes. APTOS 2019 Blindness Detection (Kaggle), 3,662 labelled fundus images, ≈ 8.9 GB at full resolution; obtained from a full-resolution Hugging Face mirror (sngsfydy/aptos). Class counts 1805/370/999/193/295.
Fits free GPU?	Yes. The exact protocol (300 $\times$ 300, batch 16, 60 epochs) takes ≈ 1.3 h on a free Colab T4. Locally (Apple M2, 8 GB unified memory) B3 at 300 px swaps heavily even at batch 8, so the local pilot uses 224 px.
Last commit / usable?	Not applicable (no code). Our implementation: PyTorch 2.14, timm 1.0.30.
Metric defined?	Yes. QWK (primary), plus accuracy, precision, recall, F1, AUC-ROC. We additionally report macro-F1, per-class precision/recall/F1, macro one-vs-rest AUC and the confusion matrix.
Split defined?	Partially. A 70/15/15 train/val/test split is stated for EyePACS only; not stated for APTOS. We use a stratified 70/15/15 split with a fixed seed.
Hyper-parameters?	Partially. Adam, $\text{lr } 10^{-3}$, weight decay 10^{-4} , gradient clipping 0.1, 60 epochs (strategy 1). Batch size, APTOS epoch count, loss function, CLAHE parameters and blur kernel are not stated.

D. Under-specified details and our choices

- Loss: not stated; softmax output implies cross-entropy, which we use.

- Gradient clipping “0.1”: interpreted as global L2-norm clipping at 0.1.
- CLAHE / blur: clip limit 2.0, 8×8 tiles on the LAB luminance channel, then a 5×5 Gaussian blur, applied identically to train, validation and test images.
- Resize: plain square resize (no border cropping), as the paper mentions only resizing.
- Augmentation: none beyond CLAHE and blur, which the paper calls “augmentation” but which act as deterministic pre-processing.
- Ensemble: strategy 2 takes the majority vote over predictions from 10 evenly spaced snapshots; strategy 1 averages the 30- and 60-epoch (here: mid- and final-epoch) snapshots.
- Reported model: the paper does not say whether the final-epoch or best-validation model is evaluated; we report both.

E. Red flags noted before reproduction

Several reported numbers are internally inconsistent, which we flag as context for the reproduction gap: (i) in Table 3 of [1], precision, recall and QWK of the B3 ensemble are all exactly 0.867, and models predicting only class 0 report a macro AUC of 0.196, below the 0.5 that a constant predictor yields; (ii) in Table 6, accuracy, precision, recall and F1 agree to within 0.01 on an imbalanced dataset; (iii) the APTOS split is not described, so overlap between near-duplicate images in train and test (a known APTOS issue) cannot be ruled out. We audit duplicates ourselves in Sec. II-B.

II. Stage 3: Reproduction Protocol

A. Data

APTOS 2019 [2] contains 3,662 labelled colour fundus photographs from Aravind Eye Hospital graded 0–4 (No DR 1805, Mild 370, Moderate 999, Severe 193, PDR 295). Images were stored at a 512px longest side (aspect ratio preserved) and resized at load time. We use a stratified 70/15/15 split with seed 42 (2,562/550/550 images).

B. Duplicate audit

Hashing the raw bytes revealed 123 groups of byte-identical images (128 redundant rows); in 30 of these groups the copies carry different grades (e.g. the same photograph labelled Moderate and PDR). Under a naive random split, 29 of the 550 test images (5.3%) have an identical copy in the training set, and 5 of them with a conflicting grade. We train on the faithful split (all 3,662 rows, as the paper presumably did) and quantify the effect of this leakage by re-scoring each model on the 521 test images that have no copy in the training set (Sec. III-E). A de-duplicated protocol (conflicting groups removed, one copy per group: 3,504 images) is implemented for Stage 5.

C. Model and training

Following [1]: an ImageNet-pretrained EfficientNet-B3 [3] (timm) with global average pooling (1,536-d) feeds a head of FC(512), Dropout(0.5), ReLU, FC(512), Dropout(0.25), ReLU and FC(5). The whole network is fine-tuned with Adam (lr 10^{-3} , weight decay 10^{-4}), global gradient-norm clipping at 0.1 and cross-entropy loss. Inputs are normalised with ImageNet statistics.

Strategy 2 (headline): CLAHE [4] on the LAB luminance channel (clip 2.0, 8×8 tiles), 5×5 Gaussian blur, square resize. Predictions are taken from 10 evenly spaced epochs and combined by majority vote. Strategy 1: plain resize to 150px; predictions from the mid-point and final epochs are averaged.

D. Compute and run configurations

Exact protocol (headline). Strategy 2 exactly as published: 300×300 inputs, batch size 16, 60 epochs, snapshots every 6 epochs. These runs used a free Google Colab NVIDIA T4 GPU (CUDA, full fp32 precision, no mixed precision), took 78s per epoch (≈ 1.3 h per run) and finished without interruption. Checkpoints were written to Google Drive after every epoch so that a disconnected session could resume.

Compute-limited pilot. Before GPU access, we trained on an Apple M2 with 8GB unified memory (PyTorch MPS backend). EfficientNet-B3 at 300px exceeded the memory budget even at batch size 8 (steps took 25–45s because of swapping), so the pilot used 224px, batch size 8 and 30 epochs (snapshots every 3 epochs), ≈ 2 h per run. The paper-setting pilot was interrupted once (disk full while writing a checkpoint) and resumed from its epoch-13 checkpoint. We keep the pilot runs because comparing them with the exact protocol isolates the effect of resolution and schedule (Sec. III-D).

All four runs share the same split (seed 42), model seed and code. Each configuration is trained once with the paper’s learning rate (10^{-3}) and once with 10^{-4} (diagnostic, Sec. III-C). The EyePACS-pretraining variant behind the paper’s 0.967 was not attempted: EyePACS is 35,126 images ($\approx 10 \times$ APTOS), which is beyond our compute budget, so our target is the paper’s ImageNet-initialised result (QWK 0.954).

E. Metrics

Following the course rules and our literature review, we report QWK [5] (primary), accuracy, macro-F1, macro one-vs-rest AUC, per-class precision/recall/F1, “minority recall” (mean recall of Severe and PDR), the number of test predictions off by two or more grades, and confusion matrices. We report the final-epoch model, the best-validation-QWK model, and both snapshot-ensemble variants.

TABLE II

Reproduced test-set results on APTOS 2019 (550 test images, faithful split). Paper: QWK 0.954 (single model, ImageNet init) and 0.967 (EyePACS init + snapshot ensemble).

Configuration	Prediction	QWK	Acc.	Macro-F1	Macro AUC	Minority recall	$ \Delta \geq 2$ errors
300 px/60 ep, lr 10^{-3} (paper)	final epoch	0.808	0.785	0.536	0.887	0.159	37
	best val QWK	0.858	0.780	0.512	0.914	0.211	31
	snapshot vote	0.846	0.796	0.527	0.923	0.205	37
	snapshot mean	0.839	0.796	0.527	0.923	0.193	38
300 px/60 ep, lr 10^{-4}	final epoch	0.858	0.809	0.610	0.899	0.388	40
	best val QWK	0.860	0.813	0.599	0.908	0.342	27
	snapshot vote	0.879	0.822	0.623	0.934	0.359	29
	snapshot mean	0.882	0.825	0.629	0.934	0.370	28
224 px/30 ep, lr 10^{-3} (M2)	final epoch	0.827	0.771	0.473	0.897	0.148	48
	best val QWK	0.827	0.771	0.473	0.897	0.148	48
	snapshot vote	0.788	0.767	0.412	0.913	0.000	50
	snapshot mean	0.791	0.769	0.416	0.913	0.000	49
224 px/30 ep, lr 10^{-4} (M2)	final epoch	0.862	0.798	0.603	0.897	0.370	30
	best val QWK	0.887	0.815	0.630	0.894	0.428	24
	snapshot vote	0.887	0.822	0.613	0.930	0.347	27
	snapshot mean	0.887	0.820	0.611	0.930	0.347	27

TABLE III

Per-class test metrics, exact paper protocol (300 px, 60 epochs, lr 10^{-3}).

Grade	Model	Prec.	Recall	F1	n
No DR	best val	0.97	0.96	0.96	271
Mild	best val	0.44	0.21	0.29	56
Moderate	best val	0.65	0.93	0.76	150
Severe	best val	0.27	0.10	0.15	29
PDR	best val	0.52	0.32	0.39	44
No DR	snapshot vote	0.94	0.97	0.96	271
Mild	snapshot vote	0.50	0.38	0.43	56
Moderate	snapshot vote	0.70	0.91	0.79	150
Severe	snapshot vote	0.00	0.00	0.00	29
PDR	snapshot vote	0.53	0.41	0.46	44

III. Stage 3: Results

A. Faithful reproduction (exact protocol)

Table II summarises the test results. With the paper’s protocol and hyper-parameters, the best-validation model (epoch 15) reaches QWK 0.858 (accuracy 0.780, macro-F1 0.512), against 0.954 reported in [1]: a gap of about 0.10 QWK. The final-epoch model is weaker (QWK 0.808). The 10-snapshot majority vote, the paper’s central contribution, gives 0.846: better than the final epoch but below the best single checkpoint.

Training was unstable (Fig. 1, Fig. 3): validation QWK fell below 0.75 at epochs 3, 5, 8, 10, 17, 22, 23 and 36 (to 0.556 at epoch 36) and recovered within one or two epochs, and the best validation score (0.895 at epoch 15) was never reached again. Test QWK of the individual snapshots (epochs 6, 12, ..., 60) was 0.830, 0.770, 0.828, 0.834, 0.803, 0.522, 0.808, 0.773, 0.781 and 0.808. The members are of uneven quality and one is very poor, so the vote cannot exceed the best member. Snapshot ensembling assumes comparably good, diverse members [6], which an

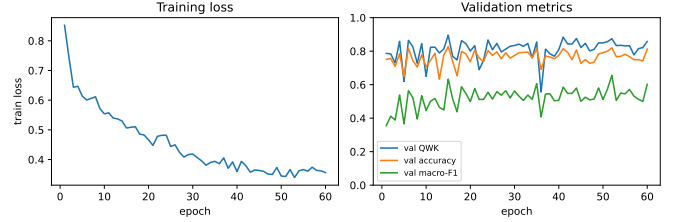


Fig. 1. Training loss and validation metrics, exact protocol with the paper’s lr 10^{-3} .

unstable trajectory without a cyclic learning-rate schedule does not provide.

B. Per-grade behaviour

QWK and accuracy hide a severe minority-grade failure (Table III, Fig. 2). No DR (recall 0.96) and Moderate (0.93) are learned well, but Severe NPDR is almost never recognised: 3 of 29 Severe eyes for the single model, 0 for the vote, and at most 5 in any snapshot. PDR recall is 0.32 (single model) and 0.41 (vote), and Mild recall only 0.21. Severe cases are mostly called Moderate (17/29) or PDR (9/29), 24 of 44 PDR eyes are called Moderate, and 34 of 56 Mild eyes are called Moderate. The errors are mostly adjacent-grade, which is why QWK stays near 0.86 even though two of five grades are essentially unrecognised; this is exactly the “accuracy (and even QWK) can be misleading” gap identified in our literature review.

C. Diagnostic: learning rate 10^{-4}

To test whether optimisation explains the gap, we repeated the exact protocol with a single change, lr $10^{-3} \rightarrow 10^{-4}$ (same split, seed, resolution, epochs and snapshots). Training became stable: validation QWK stayed between 0.81 and 0.89 for all 60 epochs with no collapses (Fig. 3),

		QWK=0.858				
True	No DR	261	8	1	0	1
	Mild	9	12	34	0	1
	Moderate	0	5	139	4	2
	Severe	0	0	17	3	9
	PDR	0	2	24	4	14
		No DR	Mild	Moderate	Severe	PDR
		Predicted				

Fig. 2. Test confusion matrix of the reproduced single model (exact protocol, paper hyper-parameters).

and training loss fell to 0.12 by epoch 10 (0.55 at 10^{-3}). The best-validation checkpoints are similar (QWK 0.860 vs. 0.858), but every other prediction improves: final epoch $0.808 \rightarrow 0.858$, snapshot vote $0.846 \rightarrow 0.879$ and snapshot mean $0.839 \rightarrow 0.882$ (macro-F1 0.629). Minority recall rises from 0.211 to 0.370 (ensemble mean; Severe 0.17, PDR 0.57).

With stable training the snapshots are uniformly good (test QWK 0.823–0.885) and the ensemble now beats the best single model, consistent with the snapshot-ensemble assumption that members are of comparable quality. Severe NPDR remains the weakest grade (5 of 29 recognised by the ensemble; 16 called Moderate) and Mild is still mostly confused with Moderate (26/56). The low training loss with a flat validation curve suggests the model overfits after about 10 epochs, which a learning-rate schedule or stronger augmentation could address. This run is the baseline we carry into Stage 4.

D. Effect of resolution and schedule

Comparing the exact protocol with our compute-limited pilot isolates the effect of using 224px for 30 epochs (Table II, Fig. 3). At 10^{-3} the full protocol helps: best single model $0.827 \rightarrow 0.858$ and vote $0.788 \rightarrow 0.846$, mostly because the vote is no longer dragged down as far by collapsed snapshots. At 10^{-4} it does not: the pilot’s best model and vote (0.887, 0.887) are at least as good as the exact protocol’s (0.860, 0.879). Resolution and schedule therefore account for at most ≈ 0.03 QWK, and the learning rate matters more than either.

E. Reproduction-gap analysis

- Duplicate leakage does not explain it. Excluding the 29 test images with an identical copy in training changes the single model’s QWK from 0.858 to 0.862 (0.879 to 0.879 for the 10^{-4} vote). The leaked images are actually predicted worse (58.6% accuracy), because several copies carry conflicting grades.

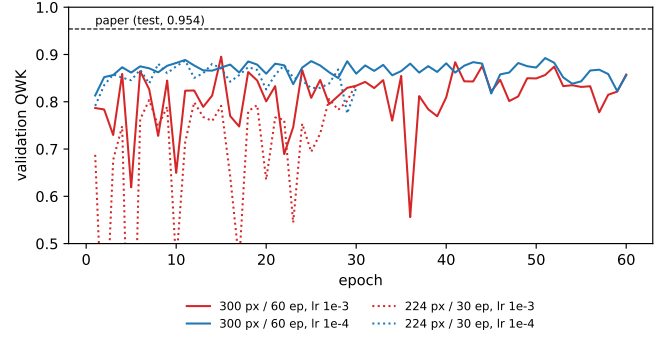


Fig. 3. Validation QWK per epoch for all four runs. 10^{-3} (red) is unstable at both resolutions; 10^{-4} (blue) is stable. The dashed line is the paper’s reported test QWK.

- Optimisation (learning rate) explains part of it. Full fine-tuning with Adam at 10^{-3} and gradient clipping at 0.1 is unstable at both resolutions and batch sizes; lowering it to 10^{-4} removes the collapses and improves the final-epoch and ensemble predictions by 0.03–0.05 QWK (Sec. III-C). The paper’s own Table 3, where VGG and ResNet models predict only class 0 under the same settings, is consistent with this.
- Resolution and schedule explain little. Moving from our 224 px / 30-epoch pilot to the exact 300 px / 60-epoch protocol adds 0.03 QWK at 10^{-3} and nothing at 10^{-4} (Sec. III-D).
- Unreported details. The split, loss, CLAHE parameters, and whether the reported model is the final or the best-validation one are not stated; each can move QWK by a few points on a 550-image test set.
- Reported numbers. The internal inconsistencies noted in Sec. 2 (e.g. identical precision, recall and QWK; macro AUC below 0.5) mean the reported 0.954 may itself not be reliable. With the stated protocol run exactly, a gap of ≈ 0.07 QWK remains that none of the stated settings account for.

F. Failure cases

Fig. 4 shows the most confidently wrong test predictions of the exact-protocol model. Eight of the ten are PDR eyes predicted as Mild or Moderate with high confidence, including one with a large pre-retinal haemorrhage; the other two are a dim, low-contrast No DR image and a Mild image both predicted as PDR. Neovascularisation is small and easy to miss, and the model appears to rely on exudates and overall image appearance instead.

IV. Limitations

Single dataset (APTOS 2019) and a single seed per configuration; no external validation; the EyePACS-pretrained configuration (0.967) was not reproduced. The 550-image test set contains only 29 Severe and 44 PDR

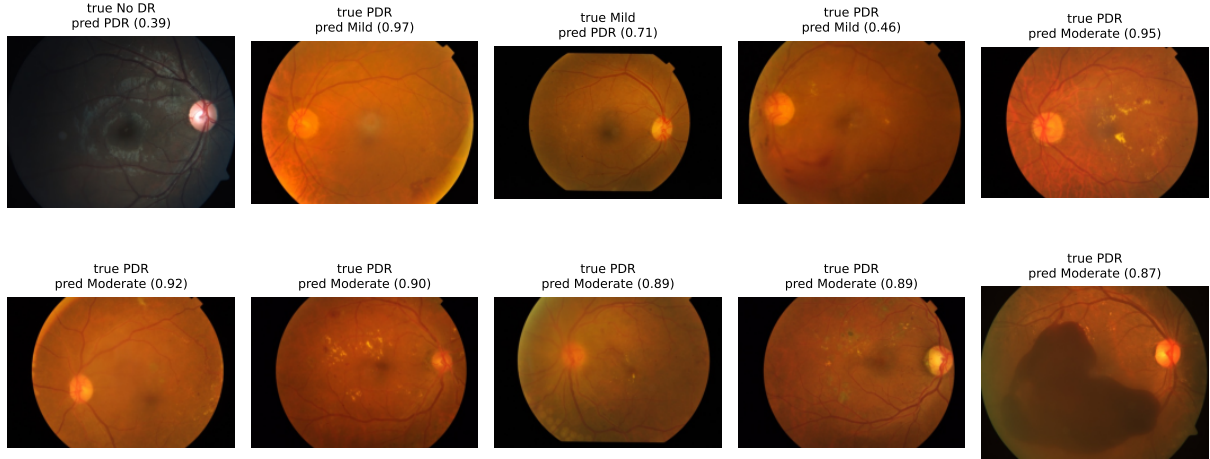


Fig. 4. Ten largest-error test predictions of the reproduced single model, exact protocol (true grade, predicted grade, confidence).

images, so per-grade recall estimates are noisy, and QWK differences of 0.01–0.02 between configurations are within run-to-run variation. This is a reproduction study; no clinical claims are made.

V. Next Step (Stage 4 preview)

The reproduction shows that the paper’s pipeline fails on exactly the grades that matter most clinically. Our Stage 4 hypothesis will therefore change one variable, the training loss, from cross-entropy to a class-balanced, ordinal-aware objective, on top of the stable $\text{lr} \cdot 10^{-4}$ exact-protocol baseline (snapshot ensemble QWK 0.882, Severe recall 0.17), and will be stated with a predicted direction and magnitude before any code is run.

References

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